

Layer 3: Transportation, EV, and V2G Mobility Report

This report provides a detailed technical breakdown of the parameters, calculations, household configurations, and modeling limitations under **Layer 3: Mobility** within the LMN Neighbourhood Design Tool.

1. Core Objectives

Layer 3 integrates electric vehicle (EV) charging demands and Vehicle-to-Grid (V2G) bi-directional discharge contribution profiles at the neighbourhood scale. It quantifies the net energy balance shift and grid stress levels under different vehicle electrification scenarios.

2. Authoritative Sizing Parameters & Sizing Formulas

Daily charging loads and discharge contributions are calculated using the following parameters and formulas from the tool's core logic:

2.1 Core Assumptions

- **EV Penetration Rate:** 1.5 EVs per household.
- **Daily Energy Demand (200 km Range):** 15.0 kWh per EV.
- **Charging Efficiency (η_{charging}):** 90% (0.90).
- **Battery Storage Efficiency (η_{battery}):** 90% (0.90).
- **V2G Participation Rate (p_{v2g}):** 50% (0.50) of the EV fleet.
- **V2G Daily Discharge ($E_{\text{v2g_per_day}}$):** 10.0 kWh per participating vehicle.

2.2 Mathematical Formulas

1. **Total EV Energy Demand ($E_{\text{EV_total}}$):** $E_{\text{EV_total}} = N_{\text{households}} \times N_{\text{EV/household}} \times \frac{E_{\text{EV_demand}}}{\eta_{\text{charging}}}$
 $E_{\text{EV_total}} = N_{\text{households}} \times 1.5 \times \frac{15.0 \text{ kWh}}{0.90} = N_{\text{households}} \times 25.0 \text{ kWh/day}$
 2. **Storage Round-Trip Loss ($E_{\text{storage_loss}}$):** Assumes 50% of the daily charging demand passes through stationary battery storage systems:
 $E_{\text{storage_loss}} = (E_{\text{EV_total}} \times 0.5) \times (1 - \eta_{\text{battery}})$
 $E_{\text{storage_loss}} = (E_{\text{EV_total}} \times 0.5) \times (1 - 0.90) = E_{\text{EV_total}} \times 0.05 = N_{\text{households}} \times 1.25 \text{ kWh/day}$
 3. **EV1 Scenario (Charging-Only):**
 - **Total EV1 Load ($E_{\text{total_EV1}}$):** $E_{\text{total_EV1}} = E_{\text{EV_total}} + E_{\text{storage_loss}} = N_{\text{households}} \times 26.25 \text{ kWh/day}$
 - **Net Energy Balance (EV1):** $E_{\text{balance_EV1}} = E_{\text{balance_base}} - E_{\text{total_EV1}}$
 4. **EV2 Scenario (V2G-Enabled):**
 - **V2G Discharge Contribution ($E_{\text{v2g_discharge}}$):**
 $E_{\text{v2g_discharge}} = N_{\text{households}} \times N_{\text{EV/household}} \times p_{\text{v2g}} \times E_{\text{v2g_per_day}}$
 $E_{\text{v2g_discharge}} = N_{\text{households}} \times 1.5 \times 0.50 \times 10.0 \text{ kWh/day} = N_{\text{households}} \times 7.5 \text{ kWh/day}$
 - **Total EV2 Load ($E_{\text{total_EV2}}$):** $E_{\text{total_EV2}} = E_{\text{total_EV1}} - E_{\text{v2g_discharge}}$
 $E_{\text{total_EV2}} = N_{\text{households}} \times (26.25 - 7.5) \text{ kWh/day} = N_{\text{households}} \times 18.75 \text{ kWh/day}$
 - **Net Energy Balance (EV2):** $E_{\text{balance_EV2}} = E_{\text{balance_base}} - E_{\text{total_EV2}}$
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3. Dwelling Unit and Household Composition

Residential household (\$HH\$) counts and gross floor areas (\$GFA\$) are determined by the building typologies composing each Neighbourhood Unit (NU) row in

Full_NUs_Archetypes.csv :

Building Typology	Households per Building	Building GFA (\$m ²)
Detached (2-Storey)	1	220
Attached (2-Storey)	1	180 (190 for MU-HS)
Mid-Rise (4-Storey)	32	3,136
High-Rise (10-Storey)	160	23,512
High-Rise (15-Storey)	240	35,268
High-Rise (20-Storey)	320	47,024

Note: Non-residential buildings (offices, retail, schools, hotels, hospitals, warehouses, datacenters, laboratories, and commercial towers) contribute 0 EV loads, resulting in zero EV demand for purely commercial or industrial NUs.

4. Grid Status & Metrics Normalisation

- **Grid Status Indicator:** Classified as "Grid Stressed - Deficit" if the net balance is negative (load > 0), and "Load Balanced - Net Zero" otherwise. Since $E_{\text{balance_base}}$ defaults to 0.0, both EV1 and EV2 represent net deficits, with EV2 reducing the magnitude of grid stress.
- **Normalised Intensity (kWh/\$m²-day):** Calculated by dividing the net energy balance (EV1/EV2) by the authoritative total floor area of the neighbourhood (\$MORPH_FLOOR\$).

5. Known Limitations

- **Static Travel Habits:** Daily vehicle mileage is modeled using a static average (15 kWh/day, equivalent to ~200 km range) rather than dynamic agent-based travel diaries or commuting variations.
- **Simplified Round-Trip Battery Loss:** Stationary battery storage loss is fixed to 50% throughput, ignoring battery chemistry, degradation over time, or temperature-dependent capacities.
- **Unconstrained V2G Discharge:** Assumes simultaneous bi-directional charging availability without local transformer constraints, grid congestion limits, or smart peak-shaving dispatch controls.
- **Baseline Decoupling:** Building-level electrical demand profiles are not integrated into the base balance ($E_{\text{balance_base}} = 0$), leaving the EV loads isolated from daily building peak load shapes.